Introducing Time Series Analysis

Topic 1

Goals

- To review basic, but crucial, statistical concepts
- To introduce what is time series analysis
- To motivate why we need special techniques for analysing time series
- To start exercising our analytical skills with algebra that steadily increases in complexity

Background knowledge

Statistics

It is assumed that you have previously studied basic probability theory and statistical inference and have a sound understanding of the following:

- Distinction between a population and a sample
- Descriptive statistics — sample mean, median, quartiles, variance
- Random variables and stochastic processes
- CDFs, PMFs, and PDFs of random variables
- Expected values and linearity of the $\mathsf{E}(\cdot)$ operator
- Population variance, covariance, and correlation
- Sampling theory, estimation, unbiasedness, and efficiency
- Mean square error and Markov/Chebyshev inequalities
- Normal distribution
- Hypothesis testing — Type I and II errors, significance level, power, p-value
- Confidence intervals
- Convergence in probability, LLN, $p$lim operator, and consistency
- Convergence in distribution, CLT, and limiting distributions

Econometrics

It is assumed that you have previously studied basic econometrics and have familiarity with:

- Multiple linear regression models — classical and generalised settings
- Estimation via OLS, MME, and MLE
- Asymptotic evaluation of estimators — consistency, asymptotic efficiency, asymptotic normality

> **!** Important
>
> - If any of these topics are unclear to you, please seek my help in office hours as early in the term as possible.
> - Alternatively, review any good introductory statistics/econometrics book of your choice. I can suggest *Larsen and Marx* and/or *Jeffrey Wooldridge* for self-study.

Basic concepts

What is a random variable?

- A random variable is a **function** $Y : \Omega \mapsto \mathbb{R}$ that maps the outcomes of an experiment to a real number.
- Random variables are denoted by capital italicised Roman letters such as $Y = Y(\omega)$, where $\omega \in \Omega$.
- For example, suppose H and T are the two possible outcomes of a coin-tossing experiment. The statement

$$Y = 1 \text{ if } H \text{ occurs and } Y = 0 \text{ if } T \text{ occurs}$$

defines a random variable.

A real-valued function of the sample space.

What is a stochastic process?

- Consider a **collection** of random variables denoted by the set

$$\{Y_i : i \in \mathbb{Z}\} \quad \text{or} \quad \{Y_i\}_{i\in\mathbb{Z}},$$

  commonly referred to as a stochastic process. The indexation i represents an arbitrary dimension along which the random variables are ordered within the collection.

- Given data $\{y_1, \dots, y_N\}$, the standard approach models the dataset as a finite-length N realisation of this stochastic process.
- The relationship among the random variables is one of its critical defining features — for instance, $\{y_1, \dots, y_N\}$ is a random sample if the random variables are iid.

A collection of random variables.

What is a time series?

- Consider what happens when we switch notation from i to t, characterising our dataset as $\{y_t\}_{t=1}^{N}$ instead of $\{y_i\}_{i=1}^{N}$.
- We now think of the subscript as denoting a point in time. The sampling dimension is now **temporal**.
- Algebraically this switch looks inconsequential. Conceptually, the change is monumental.
- If we think of the data as a collection of observations over time, three issues immediately arise that cannot be ignored from a modelling perspective.

A stochastic process indexed over time.

What key issues emerge when considering time series?

Issue 1 — Ordering. There exists a natural fixed ordering of the data. Re-shuffling, for whatever statistical reason, would not be advisable.
Issue 2 — Stability. There exists a clear logical requirement to characterise stability of the data-generating process over time. Otherwise our sample is an amalgamation of data

from distinct probabilistic regimes.

Watch: Illustration

Estimation under Instability

Issue 3 — Dependence. There exists an overriding need to manage the presence of dependence. One may be willing to assume independence over the cross-section, but one could never justify independence over time. Dependence is the defining feature of the field of time series analysis.

Issues 2 and 3 may seem abstract at present, but their meaning will become clear when we discuss stationarity and ergodicity later.

What is the defining feature of (stable) time series?

- Consider again an earlier statement: dependence is the defining feature of the field of time series analysis. What does this statement mean?
- It means that we will distinguish one time series model from another by reference precisely to their dependence characteristics.
- In particular, our primary tool will be with so-called autocovariances. These functions (of lag length) capture the linear dependence characteristics between the random variables comprising a time series.

We characterise time series via their dependence characteristics. In this course, as in most introductory courses, we will focus solely on linear dependence (autocovariances).

Illustrative examples of (stable) time series

- An *iid* process $\{Y_t\}_{t\in\mathbb{Z}}$ has no linear or non-linear dependence on the past — since the random variables are independent, the lag h autocovariance, $\gamma_Y(h) := \text{Cov}(Y_t, Y_{t+h})$, satisfies $\gamma_Y(h) = 0$ for any $h \neq 0$. We will later refer to an *iid* process as **strong white noise**.
- By contrast, consider a process where $Y_t = \varepsilon_t + 0.5\varepsilon_{t-1}$ and $\{\varepsilon_t\}$ is strong white

noise. We have

$$
\begin{aligned}
\gamma_Y(-1) &= \operatorname{Cov}(Y_t, Y_{t-1}) \\
&= \operatorname{Cov}(\varepsilon_t + 0.5\varepsilon_{t-1},\, \varepsilon_{t-1} + 0.5\varepsilon_{t-2}) \\
&= 0.5\operatorname{Var}(\varepsilon_{t-1}) \text{ since } \{\varepsilon_t\} \text{ is } iid \\
&\neq 0.
\end{aligned}
$$

Notice that $\gamma_Y(1) = \gamma_Y(-1) \neq 0$ and $\gamma_Y(h) = 0$ for all $|h| \geq 2$. We will later refer to this as a **moving average process of order 1**.

These are examples of processes with and without serial correlation, respectively.

i Watch: Live Simulation

Interactive MA(1) simulation — available in the online (HTML) version of these notes.

Students sometimes ask why I present an *iid* process as a time series example after emphasising that time series are characterised by dependence. The answer: an *iid* process indexed by time is a special case of a time series — one where the dependence happens to be zero. Whether you call it a "genuine" time series is a matter of perspective.

What is time series analysis?

- We will soon see (see "A stark message" below) that the analysis of time series data can lead to new and unique problems in statistical modelling and inference.
- Specifically, the obvious possibility of dependence introduced by the sampling of adjacent points in time can severely restrict the applicability of the many conventional statistical methods traditionally dependent on the assumption that these adjacent observations are independent and identically distributed.

The systematic approach by which one goes about answering econometric questions posed in the presence of time-ordered data exhibiting inter-temporal dependence is commonly referred to as time series analysis.

Two approaches to time series analysis exist. The **time domain** approach studies lagged relationships — e.g. how today's value affects tomorrow's. The **frequency domain** approach studies cycles by, in effect, regressing the series on sinusoids at different frequencies. This module focuses exclusively on the time domain approach.

> **i** Watch: Illustration
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> **Time vs. Frequency Domains**

Variance, covariance, and correlation

Covariances — and their time-series counterpart, autocovariances — are central to everything that follows. In this section we recap the key concepts summarising the second-order structure of the probabilistic behaviour of random variables.

Definitions

Consider univariate random variables Y_1 and Y_2. The r-th **raw moment** of Y_1 is $\mathsf{E}(Y_1^r)$ and the r-th **central moment** is $\mathsf{E}[(Y_1 - \mathsf{E}(Y_1))^r]$, for $r \in \mathbb{N}$.

$$\mathrm{Var}(Y_1) := \mathsf{E}\big[(Y_1 - \mathsf{E}(Y_1))^2\big] = \mathsf{E}(Y_1^2) - (\mathsf{E}(Y_1))^2$$

The variance is the second central moment of Y_1 — a measure of dispersion.

$$\mathrm{Cov}(Y_1, Y_2) := \mathsf{E}[(Y_1 - \mathsf{E}(Y_1))(Y_2 - \mathsf{E}(Y_2))] = \mathsf{E}(Y_1 Y_2) - \mathsf{E}(Y_1)\mathsf{E}(Y_2)$$

Note that $\mathrm{Var}(Y_1) = \mathrm{Cov}(Y_1, Y_1)$ trivially — variance is a special case of covariance where both indices are equal.

$$\mathrm{Corr}(Y_1, Y_2) := \frac{\mathrm{Cov}(Y_1, Y_2)}{\sqrt{\mathrm{Var}(Y_1)\,\mathrm{Var}(Y_2)}}$$

Correlation is a scale-free measure of the strength of linear association, and $|\mathrm{Corr}(Y_1, Y_2)| \leq 1$ by the Cauchy-Schwarz inequality.

> **What is the variance of a linear combination?**
>
> Consider random variables Y_1 and Y_2 and constants a_1 and a_2. The familiar variance rule

states

$$\operatorname{Var}(a_1 Y_1 + a_2 Y_2) = a_1^2 \operatorname{Var}(Y_1) + 2a_1 a_2 \operatorname{Cov}(Y_1, Y_2) + a_2^2 \operatorname{Var}(Y_2).$$

But if you allow me to rewrite the above as:

$$\operatorname{Var}(a_1 Y_1 + a_2 Y_2) = a_1 a_1 \operatorname{Cov}(Y_1, Y_1) + a_1 a_2 \operatorname{Cov}(Y_1, Y_2) + a_2 a_1 \operatorname{Cov}(Y_2, Y_1) + a_2 a_2 \operatorname{Cov}(Y_2, Y_2),$$

then you can see that we can also use summation notation to express the result as:

$$\operatorname{Var}(a_1 Y_1 + a_2 Y_2) = \sum_{t=1}^{2} \sum_{s=1}^{2} a_t a_s \operatorname{Cov}(Y_t, Y_s).$$

Indeed, it should come as no surprise then that

$$\operatorname{Var}\left(\sum_{t=1}^{T} a_t Y_t\right) = \sum_{t=1}^{T} \sum_{s=1}^{T} a_t a_s \operatorname{Cov}(Y_t, Y_s).$$

which is an expression we will see repeatedly.

Never omit the covariance terms!

Intuition

- Students don't always find variance/covariance manipulations intuitive. However, the same students would never make the mistake of omitting cross-terms in a squaring operation! The sum of two terms squared must yield a sum of four terms. Three terms must yield nine. Let's look at an example:

$$\begin{aligned} & a^2 + ab + ac \\ (a+b+c)^2 = & +ba + b^2 + bc \\ & +ca + cb + c^2 \end{aligned}$$

  This is nine terms, right? You would never say

$$(a+b+c)^2 = a^2 + b^2 + c^2.$$

  It just looks wrong!

- But students make this mistake all the time with variances. They say

$$\mathsf{Var}(X+Y) = \mathsf{Var}(X) + \mathsf{Var}(Y).$$

Don't forget that a variance is a squared concept (see algebra below). You cannot just assume that the cross-terms are zero unless you are told or you know explicitly that they are zero.

Algebra

- Consider $\mathsf{Var}(Z) := \mathsf{E}(Z - \mathsf{E}(Z))^2$. Now suppose $Z = X + Y$. Then,

$$
\begin{aligned}
\mathsf{Var}(Z) = \mathsf{Var}(X+Y) &= \mathsf{E}\left((X+Y) - \mathsf{E}(X+Y)\right)^2 \\
&= \mathsf{E}(X - \mathsf{E}(X) + Y - \mathsf{E}(Y))^2 \\
&= \mathsf{E}\left(\tilde{X} + \tilde{Y}\right)^2, \text{ where } \tilde{X} := X - \mathsf{E}(X) \text{ and } \tilde{Y} := Y - \mathsf{E}(Y) \\
&= \mathsf{E}\left(\tilde{X}^2 + \tilde{Y}^2 + 2\tilde{X}\tilde{Y}\right) \\
&= \mathsf{E}\left(\tilde{X}^2\right) + \mathsf{E}\left(\tilde{Y}^2\right) + 2\mathsf{E}\left(\tilde{X}\tilde{Y}\right) \\
&= \mathsf{E}(X - \mathsf{E}(X))^2 + \mathsf{E}(Y - \mathsf{E}(Y))^2 + 2\mathsf{E}((X - \mathsf{E}(X))(Y - \mathsf{E}(Y))) \\
&= \mathsf{Var}(X) + \mathsf{Var}(Y) + 2\mathsf{Cov}(X, Y).
\end{aligned}
$$

- Never omit the covariance terms!

We are now ready for the final section of Topic 1.

A stark message

What is the sample mean?

- Recall one of the most **widely-used** tools in a statistician's toolkit, $\bar{Y}_N$, the humble yet ubiquitous **sample mean** defined as

$$\bar{Y}_N = \frac{1}{N}\sum_{i=1}^{N} Y_i,$$

  where $\{Y_i\}_{i=1}^N$ denotes a set of random variables.

- Notice that N appears explicitly as a subscript in the notation for $\bar{Y}_N$ in order to explicitly emphasise the **dependence of the sample mean on the sample size**.

$(1/N)\sum Y_i$; the most widely-used statistic in econometrics.

- Just take a moment to appreciate how widespread in your education, and in your private and professional lives generally, has been the concept of a sample mean.

- It was so long ago that I cannot remember the first time I was taught to add up a set of observations and divide by their total number; and I cannot even begin to count how many times I have used this operation to get a sense of central tendency (or location) of the data-generating process behind my sample (even if I have not always used those words to describe what I was doing)!

- Have you ever wondered why we all turn so often to this statistic specifically? As opposed to say a sample mode, or a sample median, or a trimmed mean, or even just an arbitrary single observation?

- The simple answer, whether you have been taught this or not, is Chebyshev's LLN. In other words, it is because the sample mean, $\bar{Y}_N$, converges (in probability) to the population mean, $\mathsf{E}(Y)$, as $N \to \infty$. In other words, under suitable conditions, a sample moment is a consistent estimator of its population counterpart. We will study Chebyshev's LLN in a lot of detail in the slides below.

- For now, we briefly contemplate the ubiquity of the sample mean. In addition to all the applications that you can undoubtedly think of yourself, let us also consider the OLS estimator for a simple linear regression model. That is, suppose

$$Y_t = \alpha + \beta X_t + \varepsilon_t,$$

  and our sample is such that $\sum_{t=1}^T (X_t - \bar{X}_T)^2 > 0$, for $t = 1, \dots, T$. The OLS estimators are

$$(\hat{\alpha}_{OLS}, \hat{\beta}_{OLS})' = \arg\min_{a,b} \sum_{t=1}^{T} (Y_t - a - bX_t)^2\,,$$

whereby we obtain via calculus the first order conditions:

$$\sum_{t=1}^{T}(Y_t - \hat{\alpha}_{OLS} - \hat{\beta}_{OLS}X_t) = 0$$
$$\sum_{t=1}^{T} X_t(Y_t - \hat{\alpha}_{OLS} - \hat{\beta}_{OLS}X_t) = 0.$$

Solving the two equations above for our two unknowns, we obtain

$$\hat{\alpha}_{OLS} = \bar{Y}_T - \hat{\beta}_{OLS}\bar{X}_T, \text{ where}$$
$$\hat{\beta}_{OLS} = \frac{\frac{1}{T}\sum_{t=1}^{T}(X_t - \bar{X}_T)(Y_t - \bar{Y}_T)}{\frac{1}{T}\sum_{t=1}^{T}(X_t - \bar{X}_T)^2}.$$

Technically, one would also need to check second order conditions for minimisation, but that is not our current focus. Rather, our goal is to take a moment instead to appreciate how the sample mean appears repeatedly embedded throughout the OLS (and also other estimation) procedure(s).

Why is the sample mean ubiquitous?

Theorem — Chebyshev's LLN. Suppose $Y_t \overset{\text{iid}}{\sim} F_Y(\mu_Y, \sigma_Y^2)$ for $t = 1, \ldots, T$ where $0 < \sigma_Y^2 < \infty$. Then, it holds that

$$\operatorname*{plim}_{T\to\infty} \bar{Y}_T = \mu_Y.$$

Proof. [See next slide.] $\square$

- The generic "F_Y" notation denotes deliberate agnosticism about the distribution.
- The result is one of convergence in probability of $\bar{Y}_T$ to μ_Y. What does it mean? Estimator $\hat{\theta}_T$ is said to converge in probability to parameter θ if and only if there exists an arbitrarily small $e > 0$ such that

  $$\lim_{T\to\infty} \Pr\left(\left|\hat{\theta}_T - \theta\right| > e\right) = 0.$$

  We alternatively say that $\hat{\theta}_T \xrightarrow{p} \theta$ as $T \to \infty$, or that $\hat{\theta}_T$ is **consistent** for θ.

A sample moment converges in probability to its population counterpart — this is why the sample mean is ubiquitous.

Note, before proceeding, that convergence in mean-square, as defined by

$$\lim_{T\to\infty} \text{MSE}(\hat{\theta}_T) = \lim_{T\to\infty} \left(\text{Var}(\hat{\theta}_T) + \text{Bias}^2(\hat{\theta}_T)\right) = 0 \text{ given an arbitrary estimator } \hat{\theta}_T,$$

implies convergence in probability due to Chebyshev's inequality (see next slide).

What is the mechanism behind Chebyshev's LLN?

Theorem — Chebyshev's LLN. Suppose $Y_t \overset{\text{iid}}{\sim} F_Y(\mu_Y, \sigma_Y^2)$ for $t = 1, \dots, T$ where $0 < \sigma_Y^2 < \infty$. Then, it holds that

$$\operatorname*{plim}_{T\to\infty} \bar{Y}_T = \mu_Y.$$

Proof. Consider $\bar{Y}_T = (1/T)\sum_{t=1}^T Y_t$. We have, by linearity of expectations, that

$$\mathsf{E}(\bar{Y}_T) = \mathsf{E}\left(\frac{1}{T}\sum_{t=1}^T Y_t\right) = \frac{1}{T}\sum_{t=1}^T \mathsf{E}(Y_t) = \frac{T\mu_Y}{T} = \mu_Y,$$

so the estimator $\bar{Y}_T$ is unbiased for μ_Y. Further,

$$\begin{aligned}
\operatorname{Var}(\bar{Y}_T) &= \operatorname{Var}\left(\frac{1}{T}\sum_{t=1}^T Y_t\right) \\
&= \frac{1}{T^2}\sum_{t=1}^T\sum_{s=1}^T \operatorname{Cov}(Y_t, Y_s) \\
&= \frac{1}{T^2}\sum_{t=1}^T \operatorname{Cov}(Y_t, Y_t) \text{ due to independence,} \\
&= \frac{T}{T^2}\operatorname{Var}(Y_t) \text{ due to identical distributions,} \\
&= \frac{\sigma_Y^2}{T}.
\end{aligned}$$

It follows (recall what is mean-square error, or "MSE", of an estimator) that

$$\lim_{T\to\infty} \operatorname{MSE}(\bar{Y}_T) = \lim_{T\to\infty} \frac{\sigma_Y^2}{T} = 0$$

since σ_Y^2 is finite. By Chebyshev's inequality, we obtain that for any $e > 0$,

$$\Pr\big(|\bar{Y}_T - \mu_Y| > e\big) \le \frac{\sigma_Y^2}{Te^2},$$

which, together with convergence in $m.s.$, completes the proof. □
(Please work on PS1, Q2 if you cannot understand these two final steps.)

Unbiasedness plus Var($\bar{Y}_T$) $= \sigma_Y^2/T \to 0$ gives convergence in m.s., hence in probability.

Why might Chebyshev's LLN fail for time series?

Notice that we had

$$\begin{aligned}\mathrm{Var}(\bar{Y}_T) &= \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\mathrm{Cov}(Y_t, Y_s)\\ &= \frac{1}{T^2}\sum_{t=1}^{T}\mathrm{Var}(Y_t) \text{ under independence.}\end{aligned}$$

- The key point is that when we consider $\mathrm{Var}(\bar{Y}_T)$, the denominator is $O(T^2)$ but the numerator is $O(T)$ due to our explicit assumption that $\mathrm{Cov}(Y_t, Y_s) = 0$ for $t \neq s$. So $\mathrm{Var}(\bar{Y}_T)$ is $O(1/T)$. (See supporting notes below this slide for what is $O(\cdot)$ notation if you have not seen it before.)
- But what happens if we allow for an arbitrarily high degree of temporal correlation? Indeed, as you can see now, **our crucial consistency result risks falling to pieces!**
- We will return to the discussion above when we investigate the ergodic theorem.

Temporal dependence inflates $\mathrm{Var}(\bar{Y}_T)$; the numerator is no longer $O(T)$ and the LLN can fail.

What is $O(\cdot)$ notation in the slide above?

- Often, it is useful to have a measure of the order of magnitude of a particular sequence. The following definition compares the behaviour of a sequence $\{b_T\}$ with the behaviour of a power of T, say T^α, where α is chosen so that $\{b_T\}$ and $\{T^\alpha\}$ behave similarly.

Definition — Big-O notation

The sequence $\{b_T\}$ is said to be **at most of order** T^α, denoted $b_T = O(T^\alpha)$, if for a finite constant $C > 0$, there exists a finite integer T^* such that for all $T \geq T^*$, $|T^{-\alpha} b_T| \leq C$. □

Closing remarks

Big picture questions

(i) What is time series analysis?

(ii) Why do we hate time series analysis?

(iii) Why do we love time series analysis?

(iv) How do we think like time series analysts?

(v) What is some practical advice for students of time series analysis?

Regular-sized picture questions

(i) How are the tutorials going to be structured?

(ii) When are the office hours for this term?

(iii) What is the assessment criteria for this module?

(iv) Is there anything else you would like to ask?

Personal comment from Ragvir

- Finally, all, I really hope you enjoy the course. I plan not only to teach you time series but also to demonstrate throughout *how to think* like a time series analyst.
- Time series analysis is necessarily hard (for reasons I will explain) but it is an amazing field that has applications well beyond the spheres of economics and finance. This includes seismology, space travel, submarine detection, medical diagnosis, speech recognition, meteorology, and so on.
- If you have never studied it before, you may find these weeks to be a little intense. Please seek assistance from your class teacher(s) and/or myself when needed.

Review questions

1. Explain formally what is a random variable, a stochastic process, a time series, and what is meant by the analysis of time series.
2. Explain what is a population moment and what is the population variance of a random variable, and what is the covariance and/or correlation between two random variables.
3. State Chebyshev's law of large numbers ("the LLN") for an *iid* process. Prove the LLN. Give an intuitive explanation of how/why the LLN works. Explain what is the empirical relevance of the LLN.
4. Explain with explicit reference to $\text{Var}(\bar{Y}_T)$ why the LLN can potentially fail when we consider time series data.

*Separate solutions will **not** be provided (answers can be found in the lecture notes). I encourage you to use my topic review questions both to help consolidate your knowledge and as discussion prompts when you study with each other and/or you wish to interact with your teacher(s).*

Further reading

- For fundamentals, I can suggest Appendix B and Appendix C from Wooldridge's *Introductory Econometrics*. (In case it helps, I would say you can actually just pick any decent undergraduate statistics textbook of your choice.)
- I do not have any specific reading to assign for the stark message section because this section represents my own personal arrangement of the information. (In case it helps, I can suggest you review concepts such as Chebyshev's inequality, Chebyshev's LLN, orders of magnitude, convergence in mean-square, and convergence in probability in any decent undergraduate statistics textbook of your choice.)
- In general, the main textbook for this course will be Shumway and Stoffer, *Time Series Analysis and Its Applications with R Examples*, to which I will hereafter refer as "SS". For Topic 1, you can try **SS (Chapter 1.1–1.2)**; but please note that the information in the stark message section is not directly from SS.

> **i** Useful textbook
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> Given the level of detail in these lecture slides and supporting notes, I do not deem it 'compulsory' that you consult the textbook. My **course is self-contained**. Nevertheless, if (i) you wish to read further on a particular topic (e.g. if my exposition was perhaps insufficient); and/or (ii) you simply develop a love for the subject and want to pursue

advanced studies in time series analysis, then SS is a great resource.